

GraphPoint: Semantic Entity Graphs and Point Trajectories for Compositional Robot Manipulation

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Abstract—

I. INTRODUCTION

II. RELATED WORK

- A. Structured Representations for Robot Manipulation*
- B. Point-Based Robot Policies*
- C. Compositional and Long-Horizon Manipulation*

III. METHOD

- A. Overview*
- B. Task-to-Graph Construction*
- C. Role-Structured Entity Graph Encoder*
- D. Point-Trajectory Flow Policy*
- E. Subtask Progress and Long-Horizon Composition*

IV. EXPERIMENTS

- A. Experimental Setup*
- B. How Does GraphPoint Compare with Established Policies?*
- C. Does the Structured Entity Graph Enable Compositional Semantic Generalization?*
- D. Does Point-Based Action Representation Enable Cross-Embodiment Transfer?*
- E. Can Atomic Subtask Policies Compose into Long-Horizon Behaviors?*

V. CONCLUSION

REFERENCES

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